Estudo Dirigido

Comparing LLM-Based Policy Generation and Reinforcement Learning for Embodied Control

Jarne Demoen
Vrije Universiteit Brussel / Universidade Federal de Santa Catarina

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1 Introduction

Reinforcement Learning (RL) has become one of the most widely used paradigms for training agents that interact with dynamic environments. In RL, an agent learns a policy by interacting with the environment and optimizing a reward signal through trial and error. Popular deep reinforcement learning algorithms such as Proximal Policy Optimization (PPO), Soft Actor-Critic (SAC), and Deep Deterministic Policy Gradient (DDPG) have demonstrated strong performance in continuous control tasks and robotics. However, reinforcement learning typically requires a large number of interactions with the environment before a useful policy emerges. This is because learning usually starts from a random policy and gradually improves through stochastic exploration. As a result, RL can be computationally expensive and sample inefficient, especially for complex environments. Recent research has explored the use of Large Language Models (LLMs) as controllers for embodied agents (Carvalho and Nolfi 2026). Carvalho and Nolfi propose a method in which LLMs generate control strategies directly from textual descriptions of the agent, environment, and task objective. The generated policy maps continuous observation vectors to continuous action vectors and can therefore directly control the agent's actuators.

In this approach, the LLM first generates an initial control strategy using its prior knowledge and reasoning capabilities. The strategy is then iteratively refined using feedback obtained from the agent's interaction with the environment. During each iteration, the model receives sensory-motor data and reward information from the previous evaluation and proposes improvements to the controller. This iterative policy refinement framework integrates symbolic reasoning from the LLM with sub-symbolic sensory-motor data collected during interaction. Unlike many existing robotics approaches, it does not require demonstration datasets or model fine-tuning.

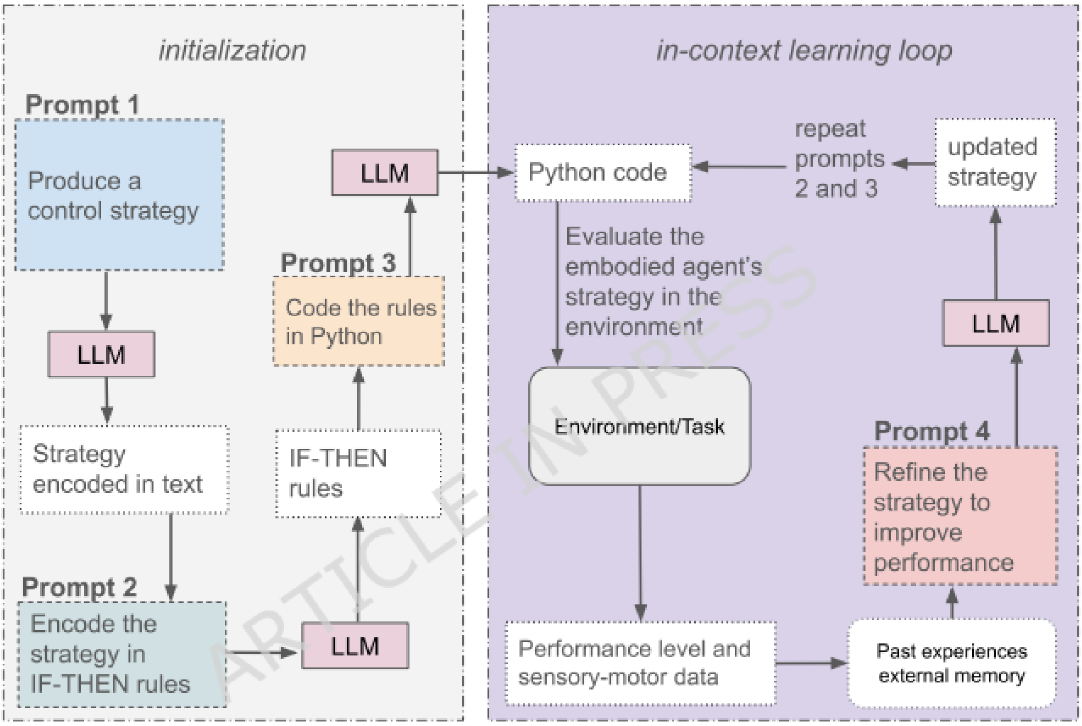

Figure 1: Overview of the prompting method enabling LLMs to directly control the actuators of embodied agents.

While this method has shown promising results, it remains unclear how it compares to traditional reinforcement learning approaches. Furthermore, recent frameworks such as the ART (Agent Reinforcement Training) system introduce automated reinforcement learning mechanisms that can train LLM-based agents using reward signals derived from task execution (*OpenPipe/ART at Auto-Rl* 2026).

2 Research Objectives

The objective of this project is to investigate whether LLM-based policy generation combined with iterative refinement can serve as a viable alternative to classical reinforcement learning methods for embodied control tasks. In particular, the project will explore whether reinforcement learning techniques applied to LLM-based agents using the ART framework can improve the performance of LLM-generated controllers. The main research questions are:

- Can LLM-generated control policies achieve performance comparable to standard (deep) reinforcement learning algorithms?
- Does applying reinforcement learning to the LLM itself using the ART framework improve the quality of generated policies?
- How do the different approaches compare in terms of learning speed, sample efficiency, and policy robustness?

3 Methodology

This study will compare three approaches to learning control policies for embodied agents.

3.1 Standard Reinforcement Learning Baselines

Several commonly used deep reinforcement learning algorithms will be implemented as baselines. These algorithms learn policies directly from interaction data using gradient-based optimization. The algorithms considered in this project include:

- Proximal Policy Optimization (PPO)
- Soft Actor-Critic (SAC)
- Deep Deterministic Policy Gradient (DDPG)

These algorithms will be trained on standard continuous control environments and serve as reference methods for comparison.

3.2 LLM-Based Iterative Policy Refinement

The second approach follows the methodology proposed by Carvalho and Nolfi (2026). In this framework, an LLM generates an initial control strategy based on a textual description of the agent, environment, and task objective. The generated strategy is converted into executable code representing the controller. The controller is then evaluated in the environment, producing sensory-motor data and reward signals. During each iteration of the learning loop, the LLM receives:

- The current control strategy
- The best strategy discovered so far
- Performance scores from previous evaluations
- Sensory-motor data collected during the most recent episode

Using this information, the LLM proposes modifications to the policy in order to improve performance. This process is repeated for multiple iterations until satisfactory performance is achieved.

3.3 LLM-Based Control with ART Reinforcement Training

The third approach extends the previous method by incorporating reinforcement learning using the ART (Agent Reinforcement Training) framework. ART allows LLM-based agents to be trained using reward signals derived from task performance. Instead of relying solely on iterative prompting, the model can be further improved by updating its parameters based on rewarded trajectories generated during interaction. In this setup, the LLM will generate policies and interact with the environment as before, but the ART framework will collect trajectories and use them to train the model through reinforcement learning updates. This allows the LLM to adapt its behavior based on accumulated experience rather than relying exclusively on prompt-based reasoning.

4 Experimental Setup

Experiments will be conducted using simulation environments from the Gymnasium library. These environments provide standardized benchmarks for evaluating control policies. Candidate environments include:

- CartPole
- Pendulum
- MountainCarContinuous
- Acrobot
- Ant

Each approach will be evaluated using identical environments and comparable computational resources. Performance will be evaluated using the following metrics:

- Average cumulative reward
- Learning speed (episodes required to reach a performance threshold)
- Policy robustness across different initial conditions
- Sample efficiency (number of interactions required for learning)

Multiple experimental runs with different random seeds will be performed to ensure statistical reliability.

5 Expected Contributions

The expected contributions of this project are:

- An implementation of LLM-based policy generation for embodied control tasks.
- A comparative evaluation between traditional reinforcement learning methods and LLM-based controllers.
- An investigation of whether reinforcement learning applied to LLM agents using ART improves controller performance.
- Insights into the advantages and limitations of reasoning-based policy generation compared to gradient-based learning.

6 Conclusion

This project explores a novel paradigm for training autonomous agents by combining the reasoning capabilities of large language models with reinforcement learning techniques. By comparing classical reinforcement learning algorithms with LLM-generated policies and ART-enhanced LLM agents, the study aims to evaluate whether LLM-based approaches can provide faster policy initialization and improved learning efficiency in embodied control tasks.

References

Carvalho, Jonata Tyska and Stefano Nolfi (Mar. 2026). "Sensory-Motor Control with Large Language Models via Iterative Policy Refinement". In: *Scientific Reports*. ISSN: 2045-2322. DOI: 10.1038/s41598-026-42091-0. (Visited on 03/29/2026).

OpenPipe/ART at Auto-Rl (2026). https://github.com/OpenPipe/ART/tree/auto-rl. (Visited on 03/29/2026).

Supervisor Approval

I, Prof. Jônata Tyska Carvalho, approve this proposal for the *Estudo Dirigido* course and agree to supervise the proposed project.

Supervisor: Prof. Jônata Tyska Carvalho

Signature:____________________

Date:____________________

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Documento assinado digitalmente
Jonata Tyska Carvalho
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